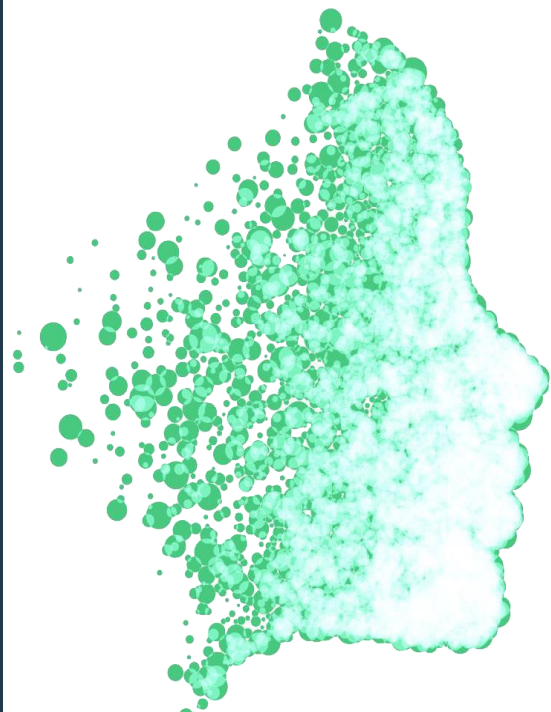




*I have a dream*



# Connected and Autonomous Vehicles

---

PhD course

© 2026 HiPeRT Lab + UPenn



**UNIMORE**  
UNIVERSITÀ DEGLI STUDI DI  
MODENA E REGGIO EMILIA

High Performance  
Real Time **Lab**



# Course outline

## Lesson #1

- › Intro course + basics of AD
- › Perception and AI
- › Hardware platforms

## Lesson #2

- › Cars and the city
- › Let's build the car
- › Navigation: planning and control

## Assignments





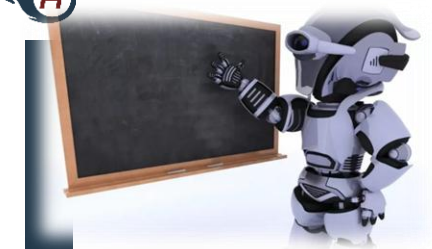
Autonomous driving

&

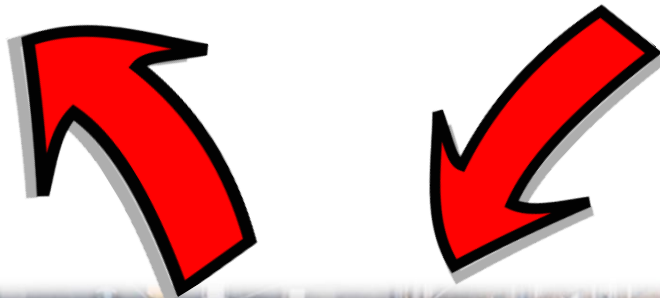
Smart cities



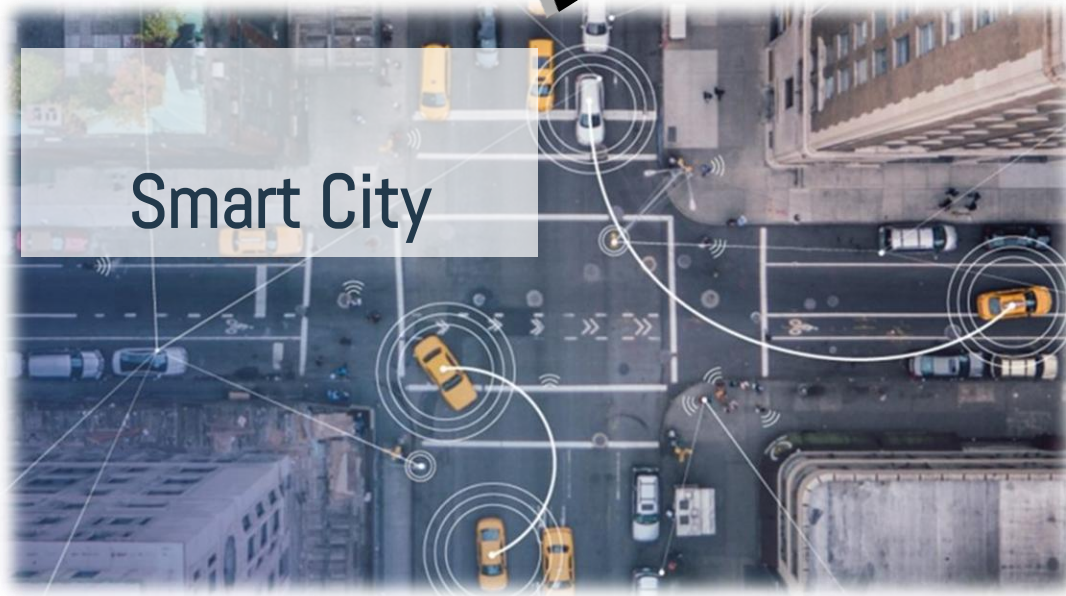
1. Train our AI



2. Additional eyes  
for the car!!



Smart City



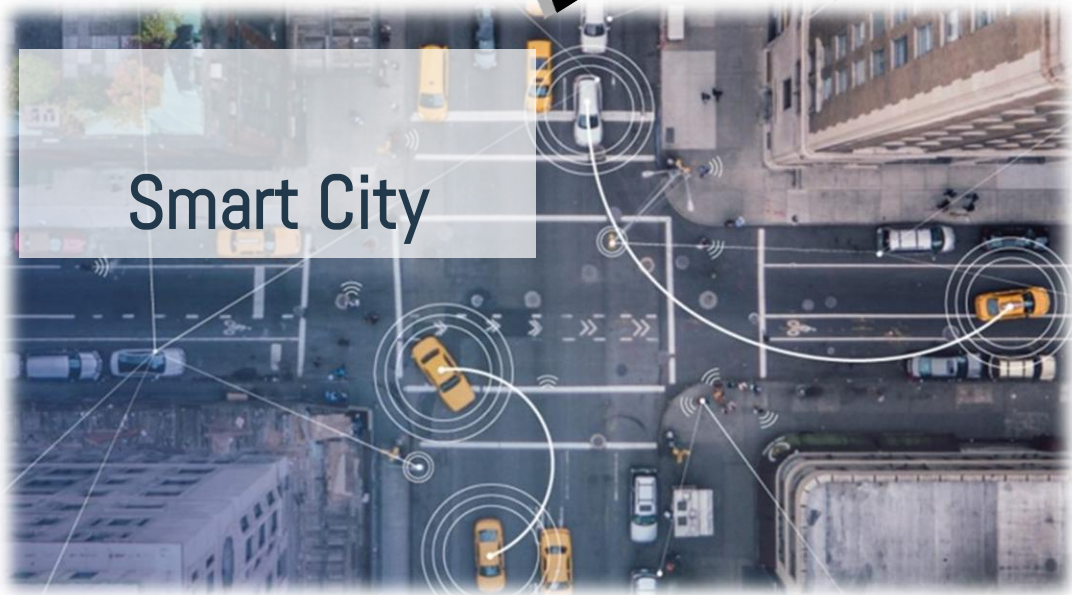


- ✓ Raw data for training sets
- ✓ Privacy and storage issues
- ✓ Real-time processing

## 1. Train our AI



## 2. Additional eyes for the car!!



- ✓ City cameras, other vehicles, pedestrians, traffic lights, traffic signs....
- ✓ Internet-of-Vehicles
- ✓ Stable 4-5G network

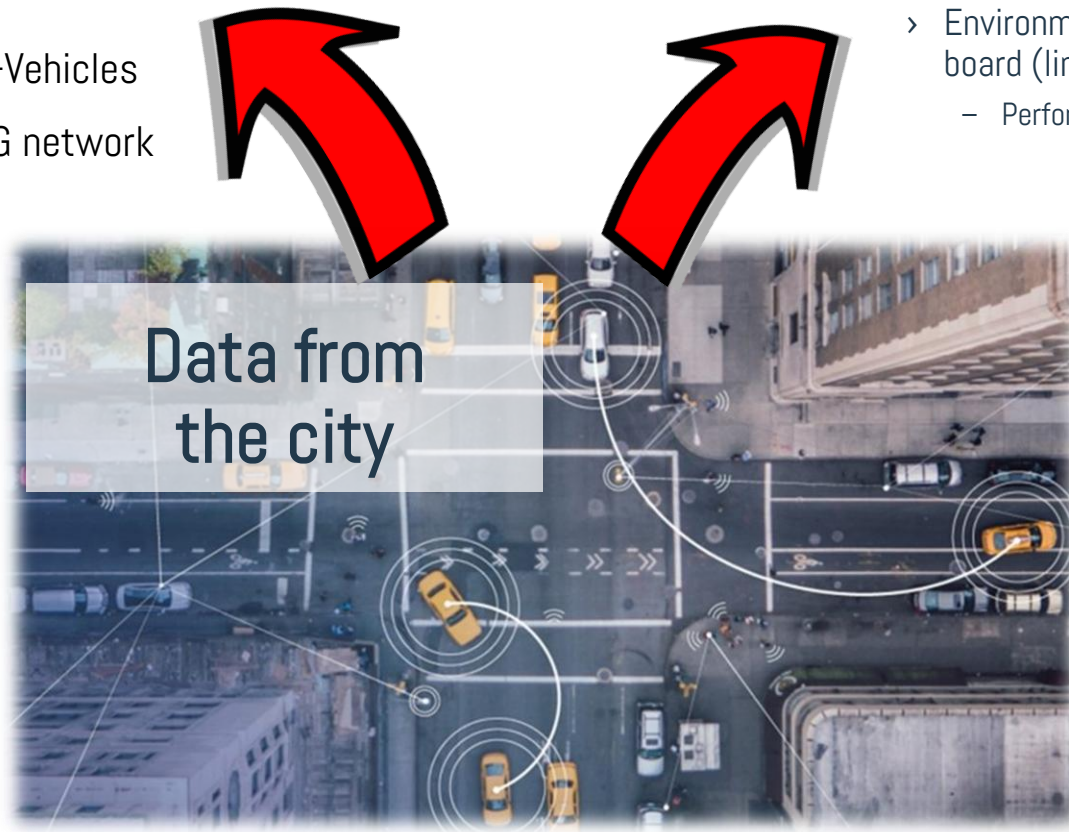


# City IoT

- ✓ City cameras, other vehicles, pedestrians, traffic lights, traffic signs....
- ✓ Internet-of-Vehicles
- ✓ Stable 4-5G network

## We use

- › Raw data for training sets
  - Privacy and storage issues
- › Environment information to enhance on-board (limited) perception
  - Performance / real time



HOME

COS'È MASA ▾

SERVIZI ▾

COMUNICAZIONE ▾

CONTATTI

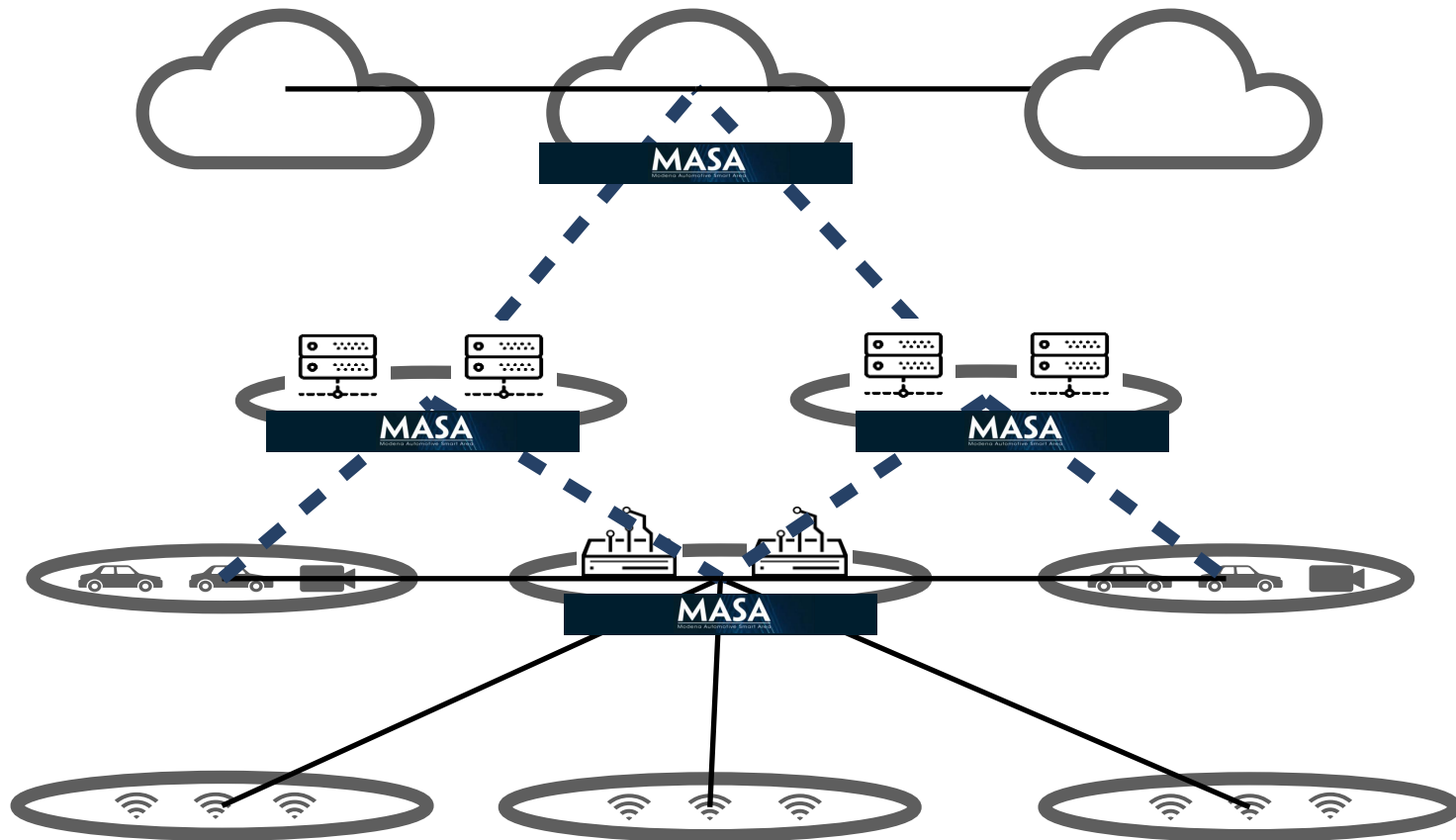
🇮🇹 ▾

# MASA

Modena Automotive Smart Area

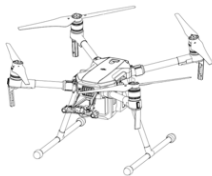
A LIVING LAB FOR AUTOMATED DRIVING

MASA è il primo laboratorio "a cielo aperto" per la sperimentazione e la certificazione delle tecnologie di guida autonoma e di guida connessa (veicoli, componenti e servizi) che interagiscono con altri veicoli, con le infrastrutture e con gli elementi dello spazio urbano.





# What, where, how?



## On field (aka: on-edge)

- › Perception; log and analyze
- › Short term sense, predict & react in RT



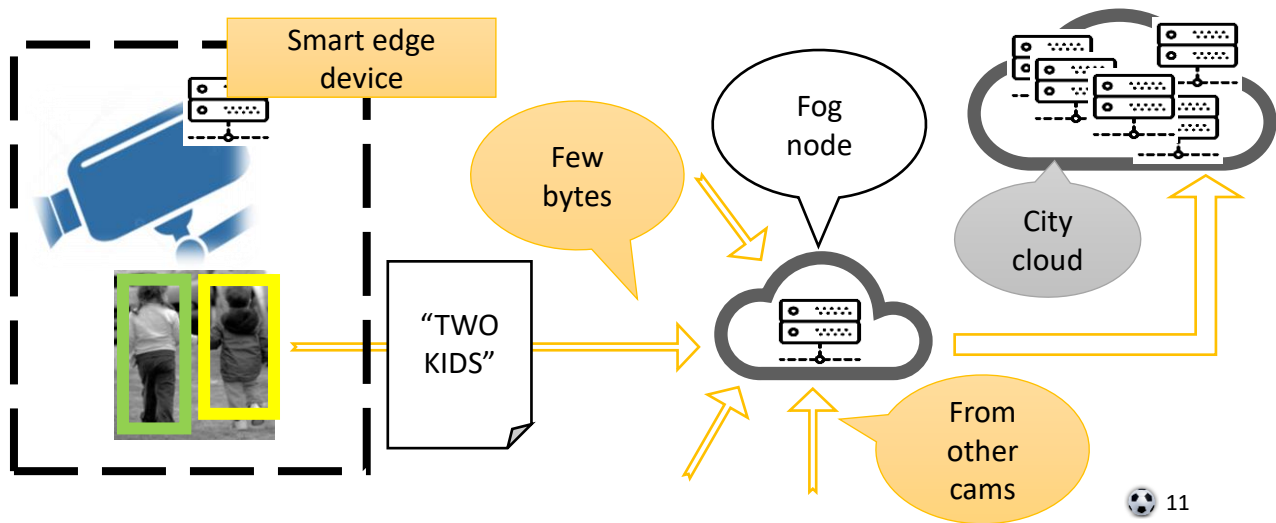
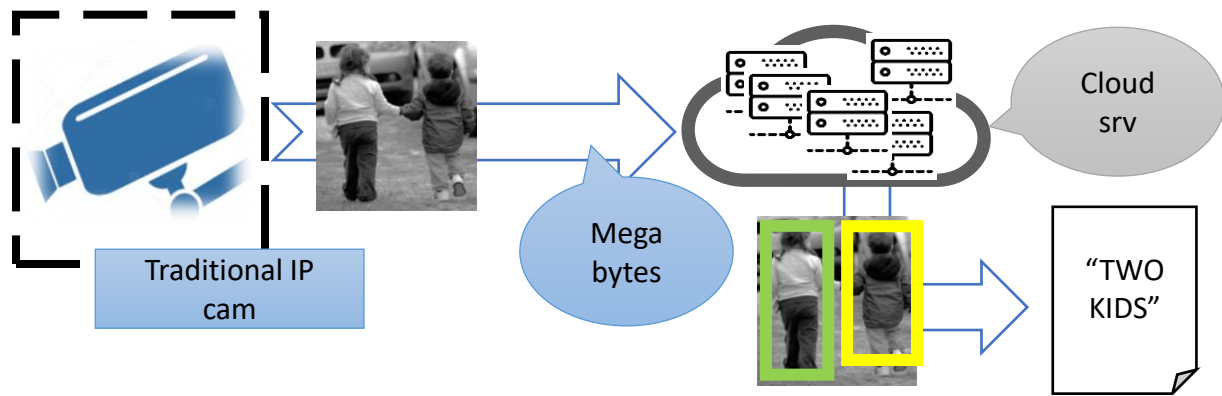
## In cloud

- › Aggregate perception from multiple edges
- › Big data analytics and logging
- › Long-term “holistic” prediction



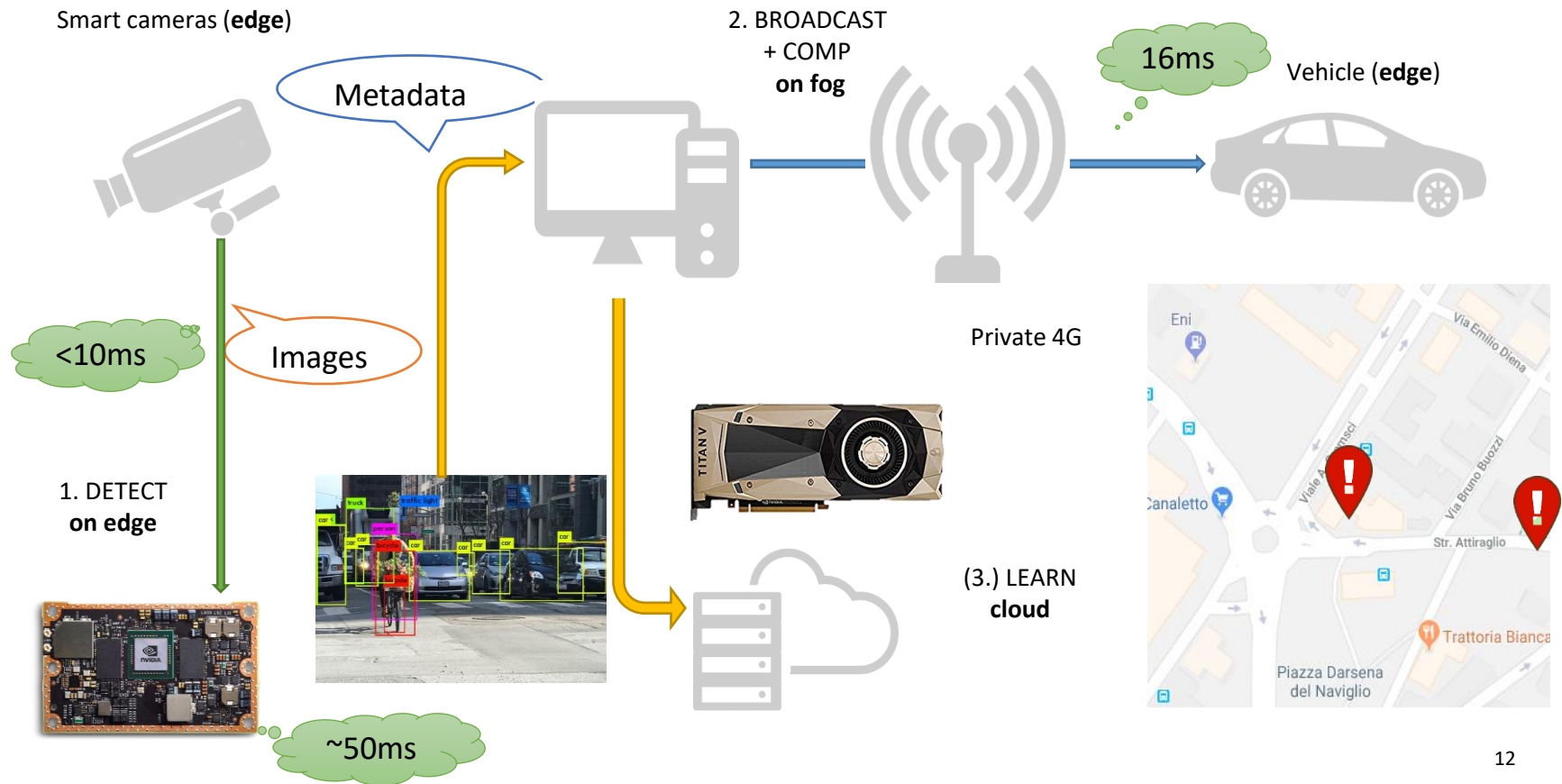
## Comms

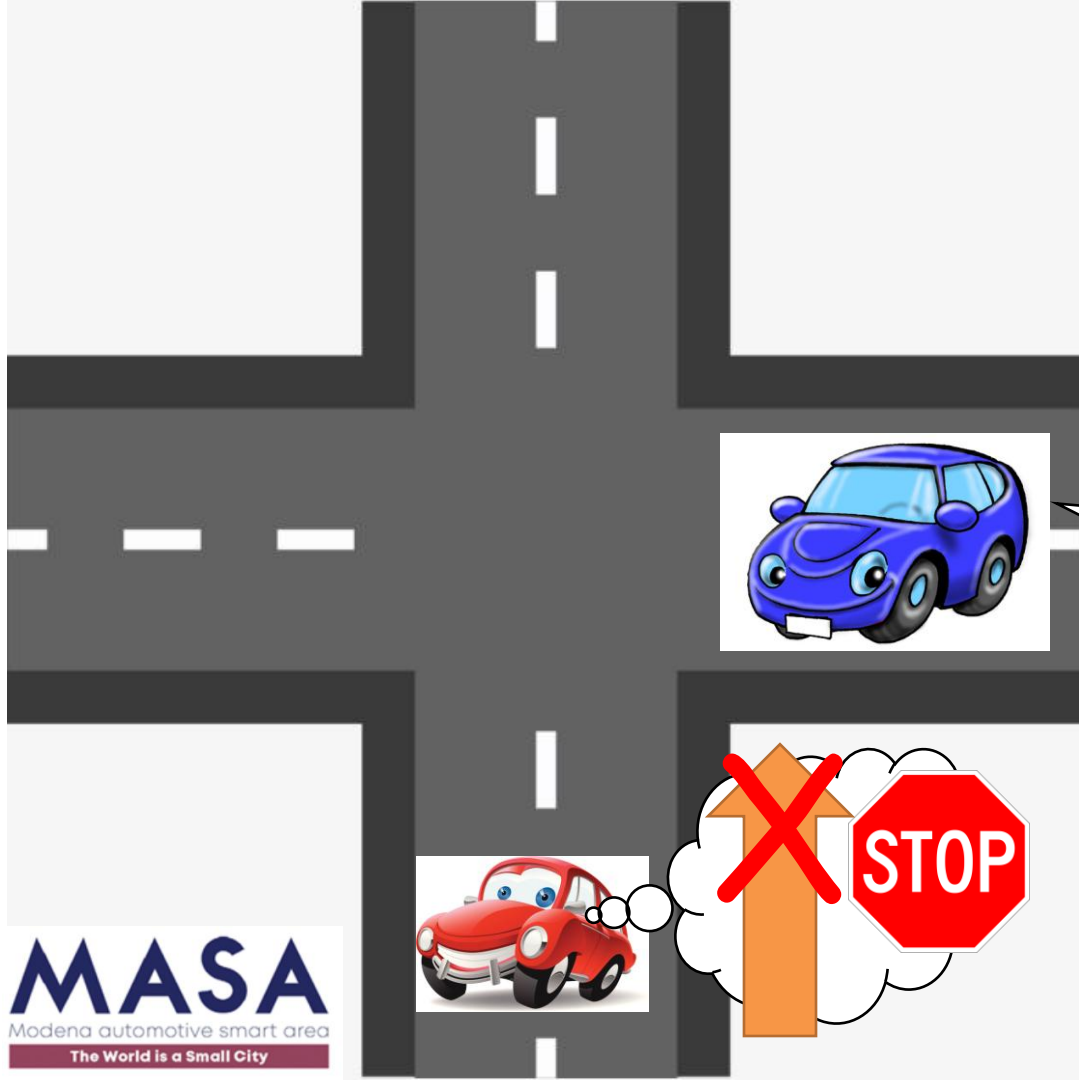
- › Safe, reliable and secure





# Fast Smart City flow: under 100 *ms*





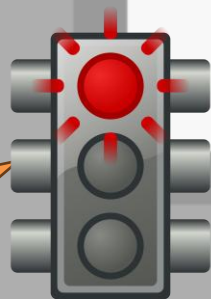


Cars speak to/from the city





-3 seconds  
to green

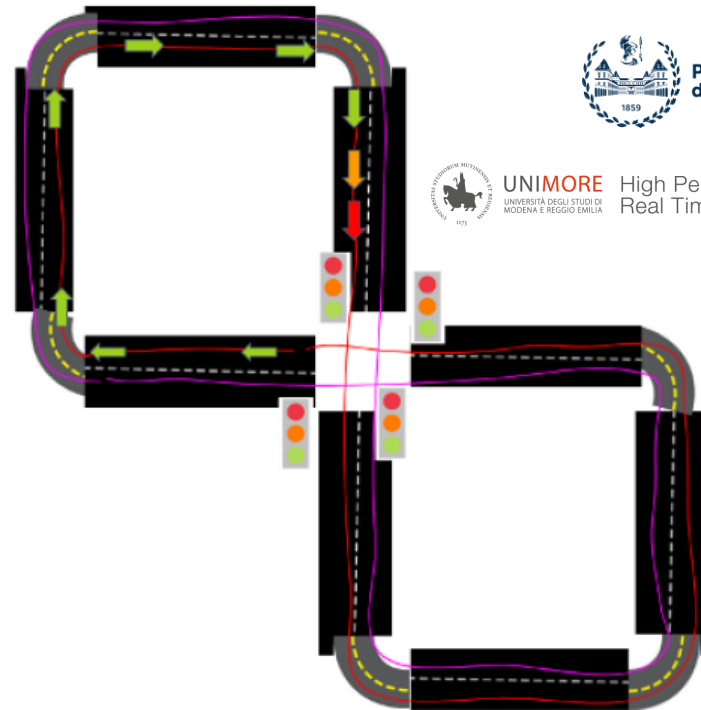




# The MoVeOver project

- › Funded by the PNRR (Funding for the Italian resiliency)
- › Target: smart intersections for Connected Autonomous Vehicles
- › Traditionally prototyped in simulators / in Matlab...

Now, let's create it!



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DELL'AQUILA



ALMA MATER STUDIORUM  
UNIVERSITÀ DI BOLOGNA



Politecnico  
di Torino



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Finanziato  
dall'Unione europea  
NextGenerationEU



Ministero  
dell'Università  
e della Ricerca



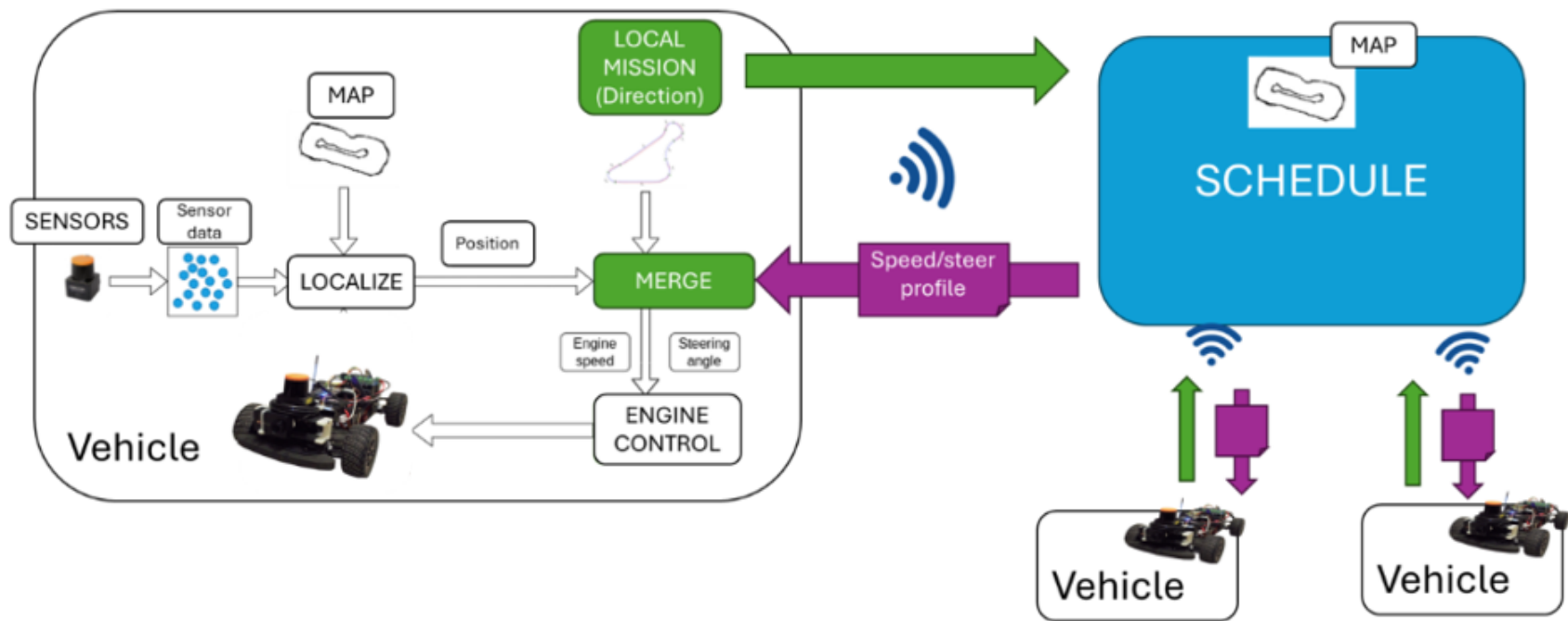
Italiadomani  
PIANO NAZIONALE  
DI RIPRESA E RESILIENZA

RESTART



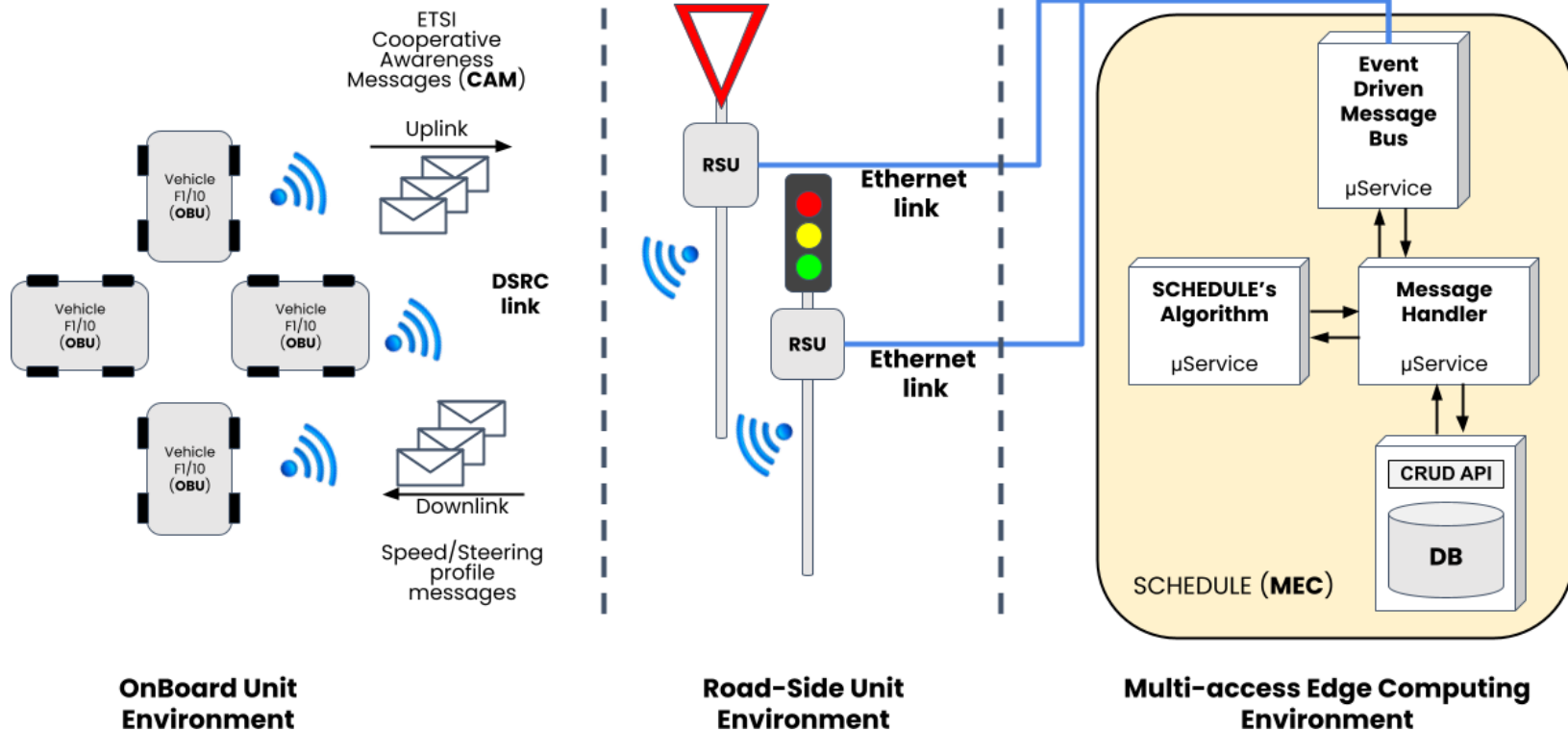
# System architecture

- › The decision module internally we call "SCHEDULE"

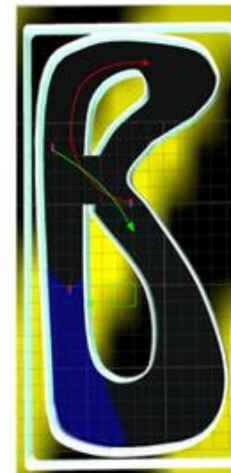
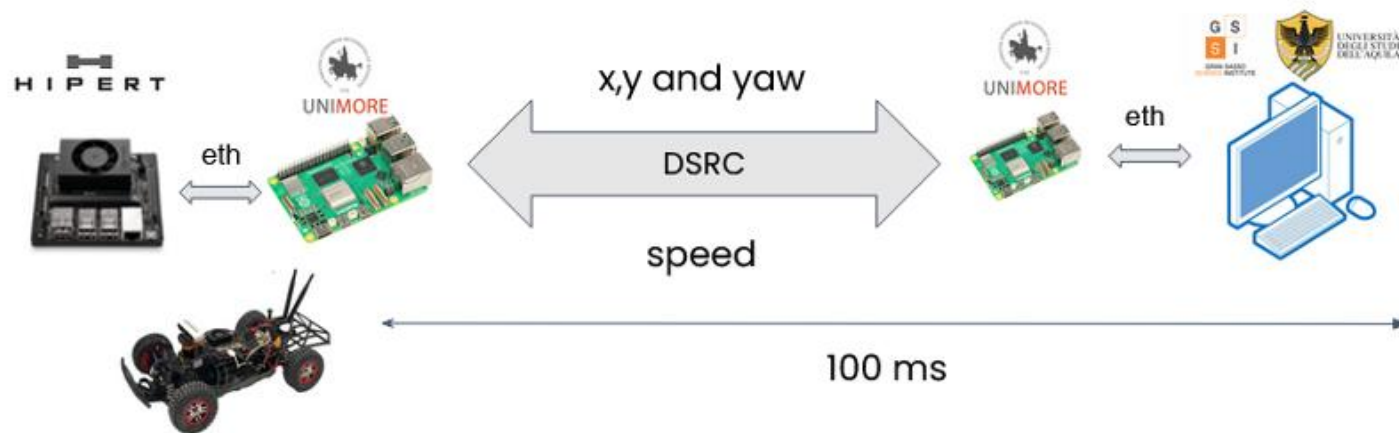




# Communication infrastructure

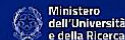


# Communication infrastructure – on-vehicle





# Not only a toy...



## RESTART Final Plenary Dissemination Event

"Shaping Horizons In Future Telecommunications" | January 19-21, 2026 | Rome

THE NEW APPLICATIONS | BEST DEMO  
**F45.1** Mini-cars demonstration of  
project traffic control  
algorithms

RESTART Scientific Coordinator  
Prof. Antonio Capone

RESTART Program Coordinator  
Prof. Nicola Blefari Melazzi

"Shaping Horizons In Future Telecommunications" | January 19-21, 2026 | Rome





# How are cars built?

Next-generation ECUs

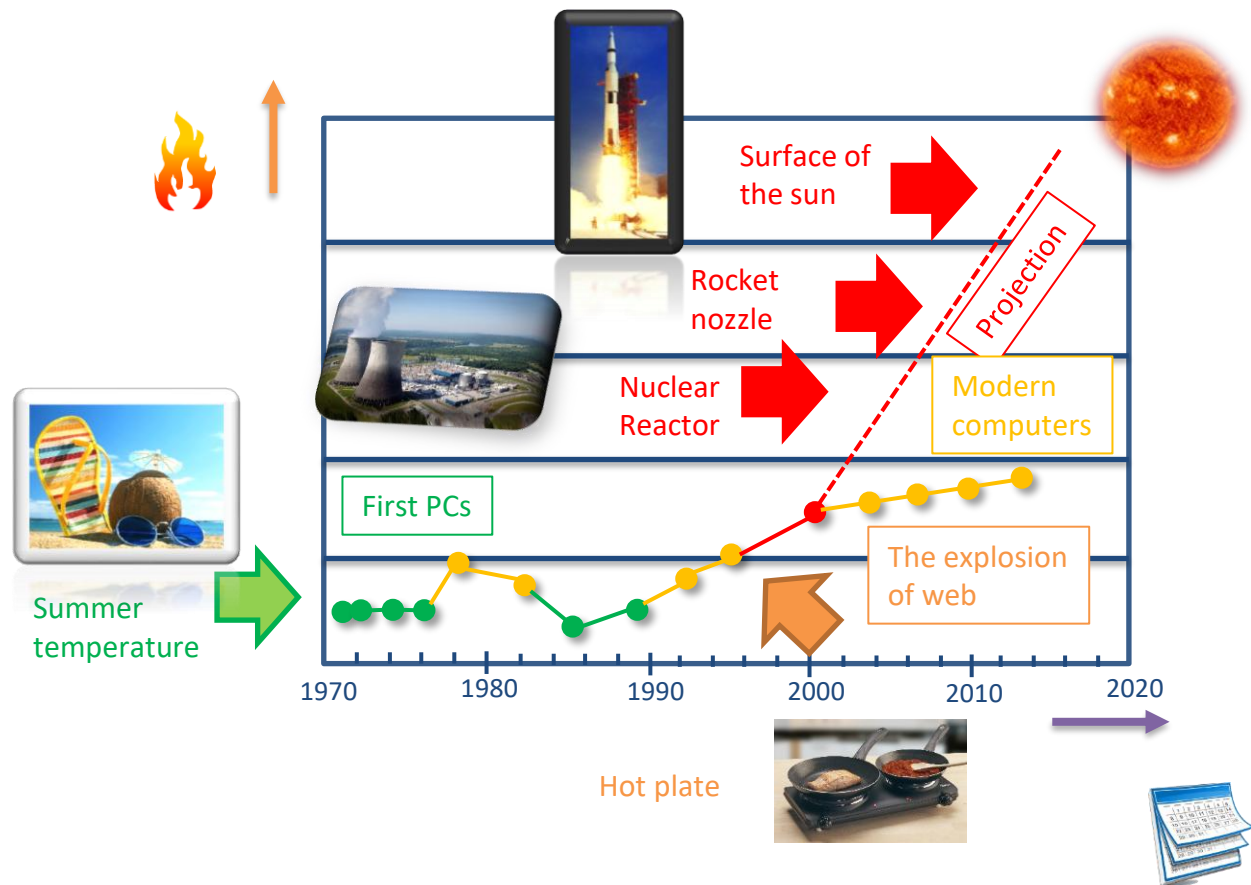


"What's the problem?"



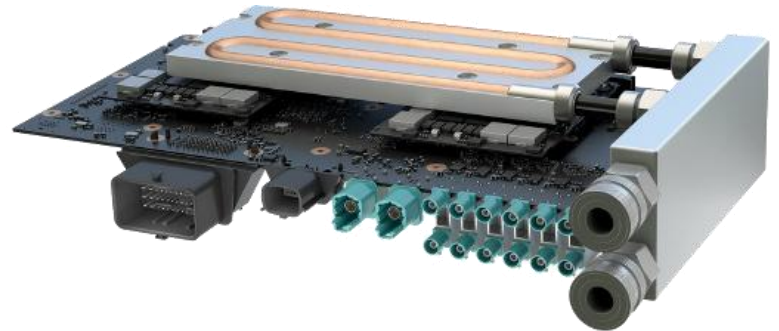
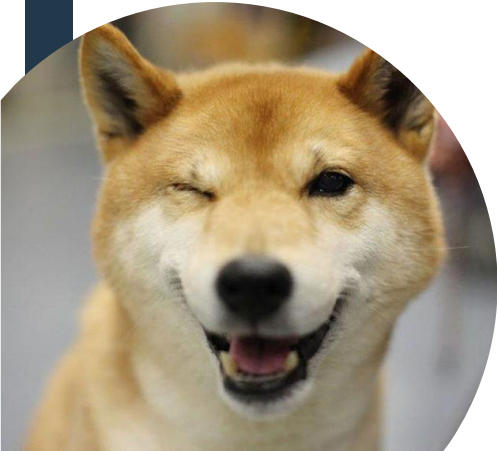


I'm hot, baby!



# A new generation of computers

High-Performance embedded systems (est. 2014-16)

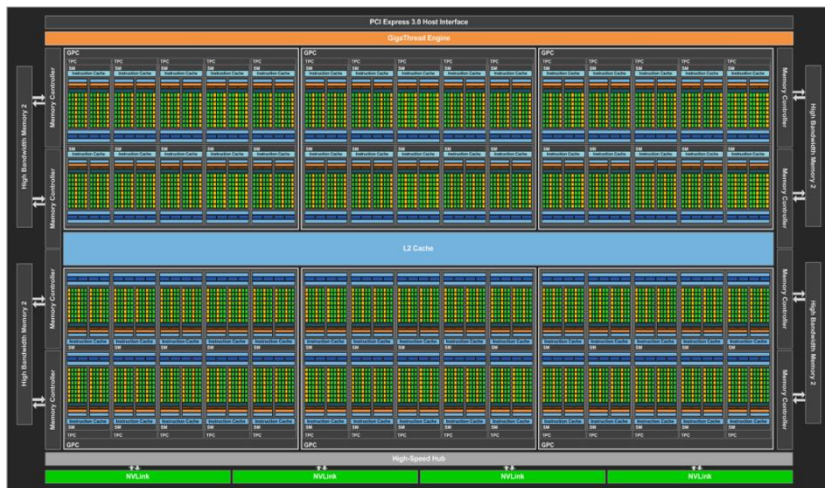




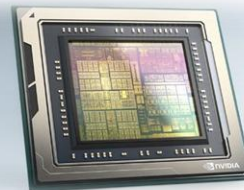
# Heterogeneous computing architectures



CPU  
MULTIPLE CORES



GPU  
THOUSANDS OF CORES



E.g.: Nvidia Orin  
200 TOPS at 40W



IBM Blue Gene (2007): 131,000 CPUs, 250 m<sup>2</sup>, 1+MW (+cooling)



# Heterogeneous computing architectures

Couple a General-Purpose Host....

- › E.g., ARM, HiSilicon
- › Typically run GNU/Linux

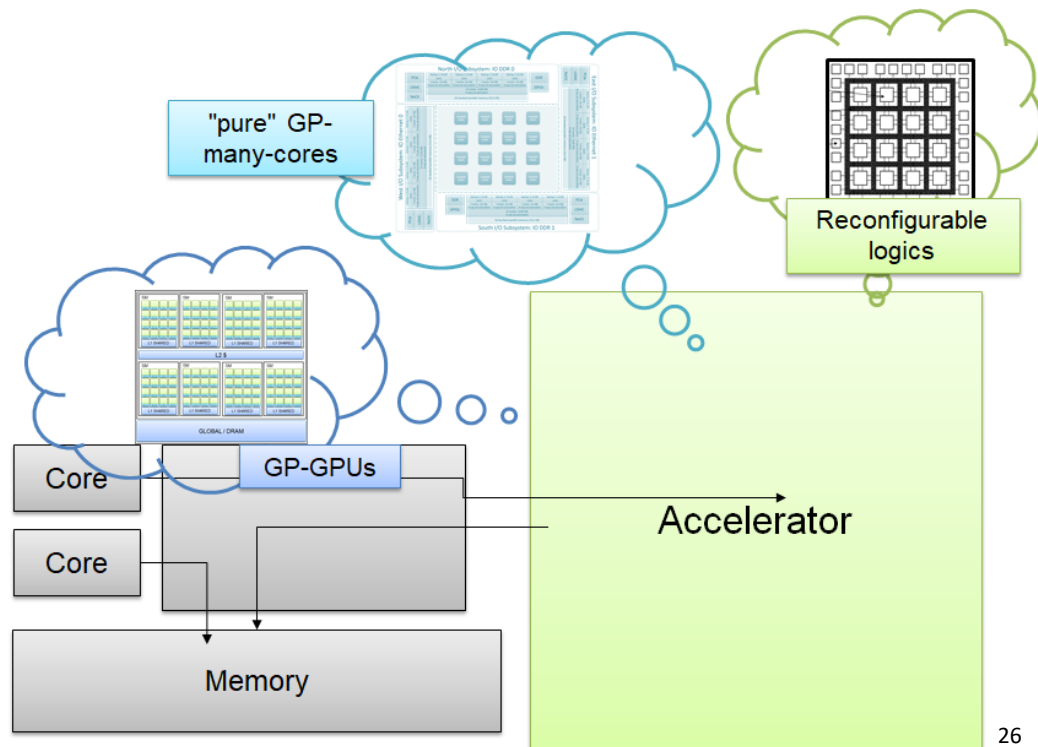
...and 1+ data cruncher accelerator(s)

- › GPGPUs vs FPGAs vs ASIC vs TPUs vs DLEs vs....

Key requirements

- › < 30/50W
- › > 2TFLOPS
- › (T)FLOPS/W is the metric

*SWaP – Size, Weight and Power*





One step back...



# Neural networks

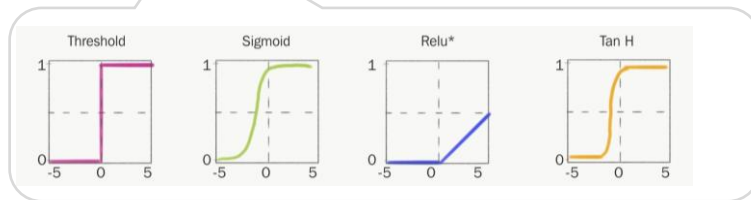
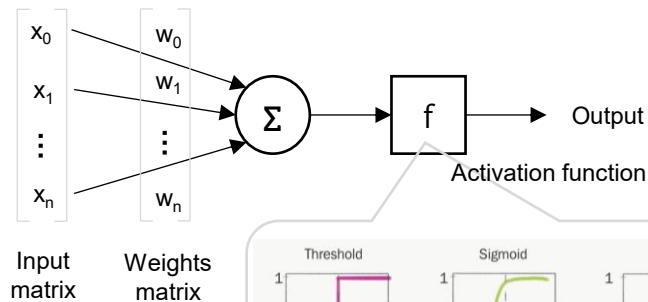
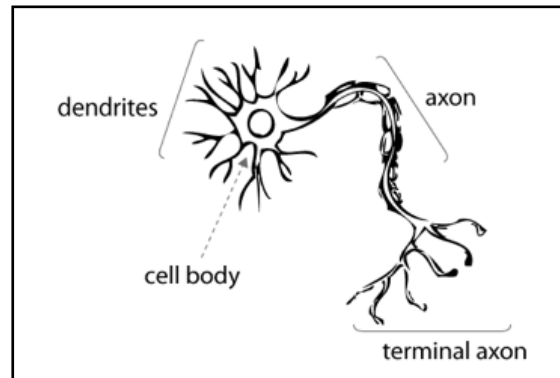
Bio-inspired

Based on neurons arranged in layers

- › And sub-layers

Convolutional neural network

- › Convolutions are widely used in computer graphics (keep this in mind..)

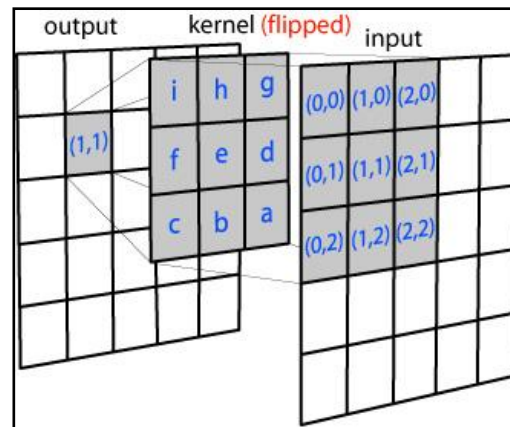




# Convolutions

- › Computation-intensive
- › Suitable for implementation in hardware
- › In computer vision, blurring

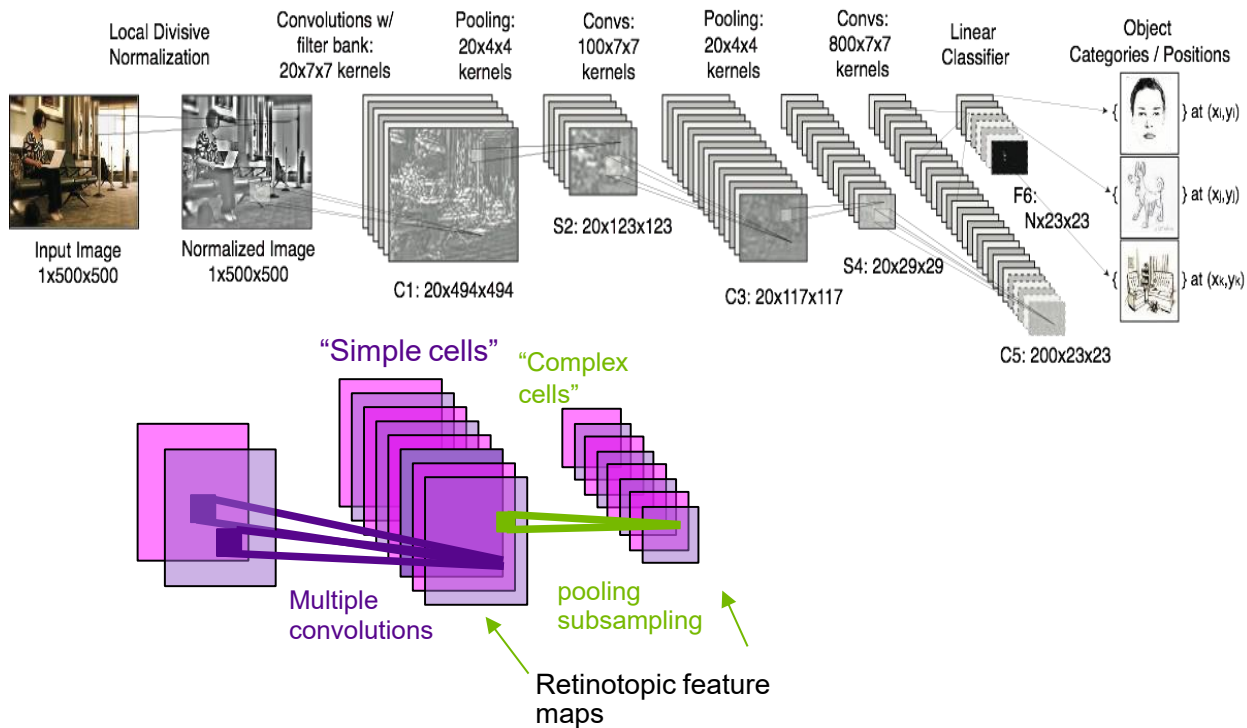
$$\begin{aligned}(f * g)(t) &\stackrel{\text{def}}{=} \int_{-\infty}^{\infty} f(\tau)g(t - \tau) d\tau \\ &= \int_{-\infty}^{\infty} f(t - \tau)g(\tau) d\tau.\end{aligned}$$





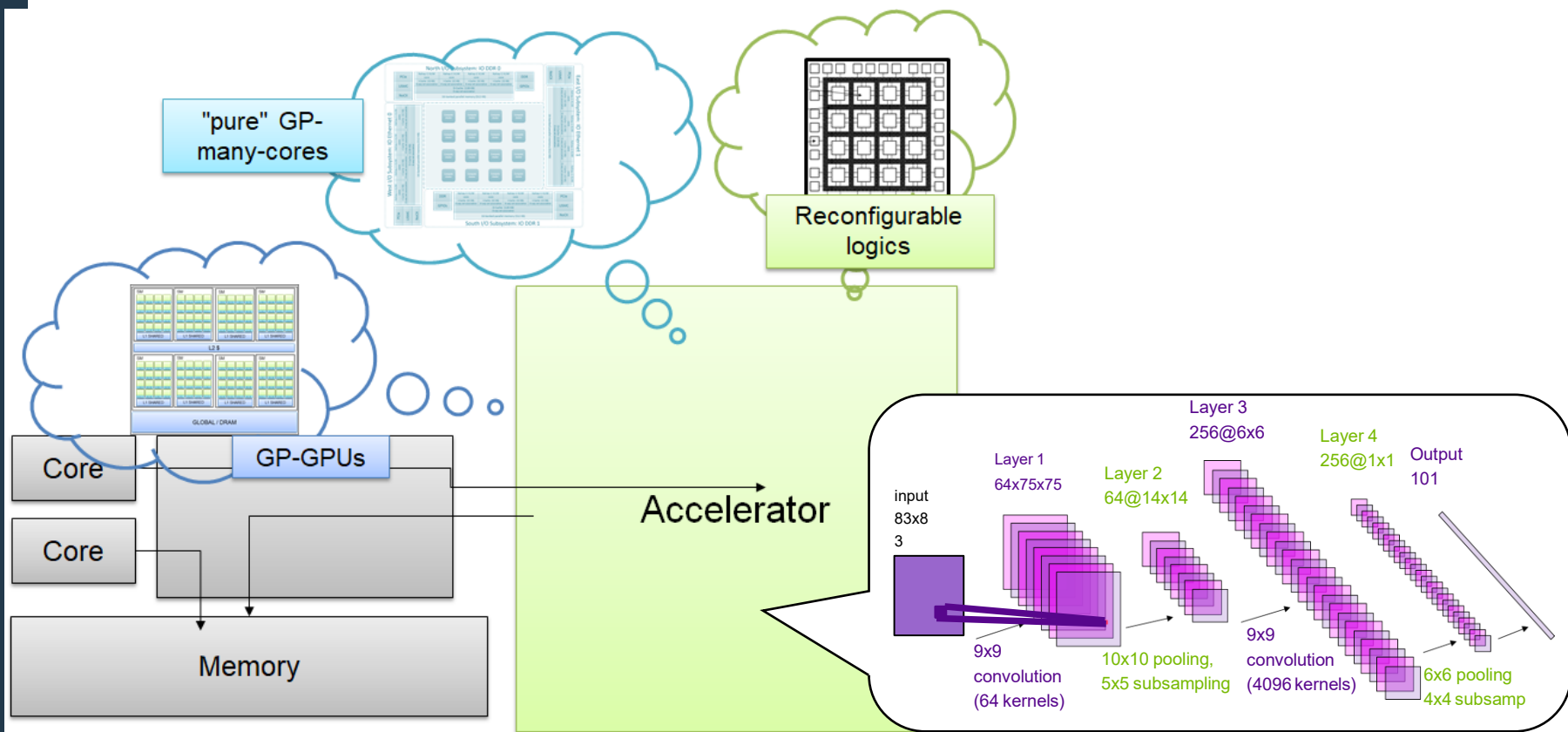
# The convolutional net model - CNN

> (Multistage Hubel-Wiesel system)





# How to deploy a neural network





# Embedded computers

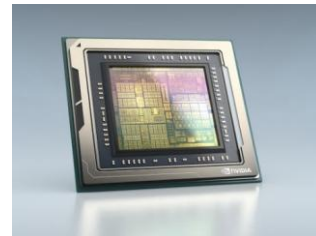
2007:

IBM Blue Gene  
200+ TFLOPS

131,000 CPUs

250 m2

1+MW (+ cooling)

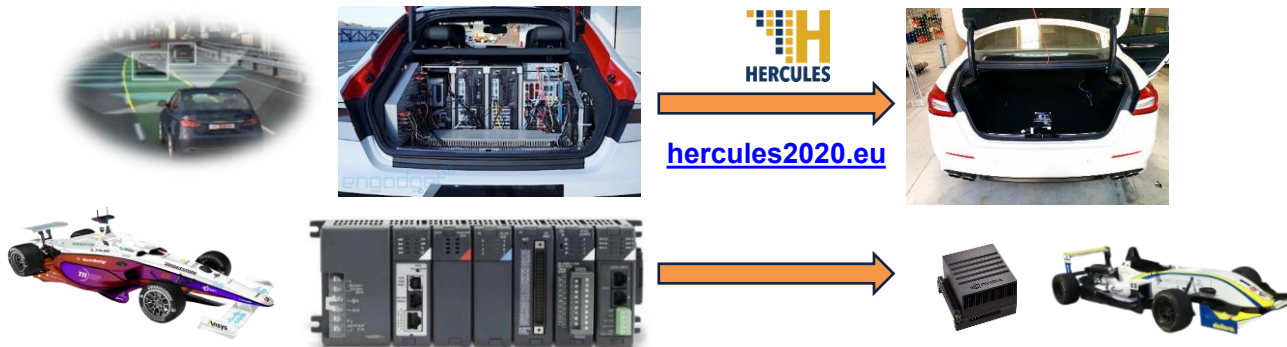


2022:

Nvidia Orin  
200+ TOPS

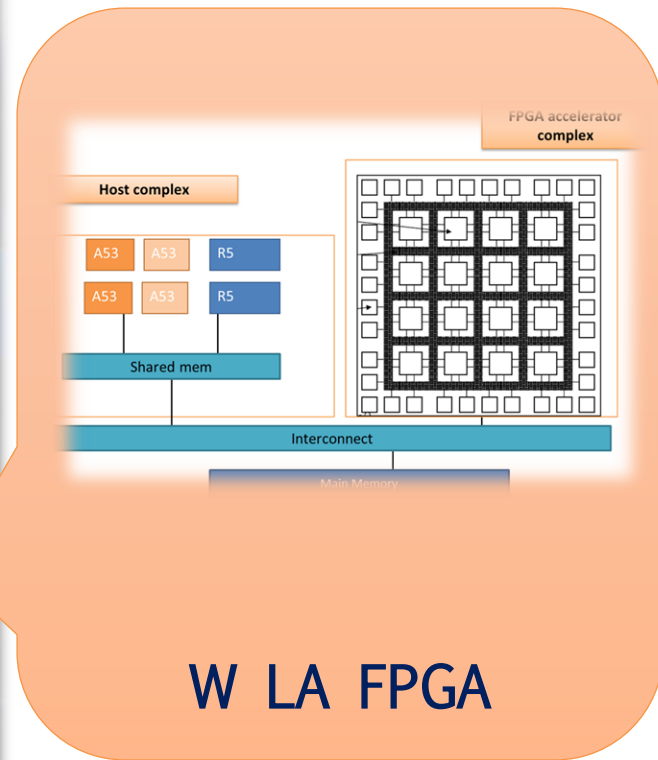
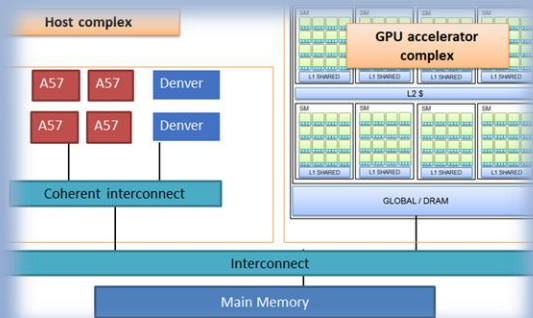
40W

Reduced Size, Weight & Power





# GPGPU SKY HARD



W LA FPGA



# Real-Time Embedded Systems

Cars are constrained!

- › Size, Weight and Power – SWaP
- › High workload (e.g, Machine Learning), require high-performance and power efficiency
- › Example: How does it take to turn on MR-23DL (“Diletta”)?

Safety critical systems require predictable performance

- › At all the level of the software stack
- › Not only application...but also RTOSes, hypervisors...

# Three dimensions

## Workload-intensive complex tasks

- › Data crunching
- › Perception, AI, but also vehicle models and linear/non linear control
- › Example: How does it take to turn on MR-23DL ("Diletta")?

## Latency-critical control tasks

- › Cyber-physical interaction
- › Tight control loops (e.g., steering PID)

## Safety critical systems require **predictable performance**

- › To be certifiable
- › Full-stack! From HW to OS to SW...



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# References



## Course website

- › <https://github.com/HiPeRT/F1tenth-PhD>

## My contacts

- › [paolo.burgio@unimore.it](mailto:paolo.burgio@unimore.it)
- › <http://hipert.mat.unimore.it/people/paolob/>

## Resources

- › <https://f1tenth.org>
- › A "small blog"
  - <http://www.google.com>